Supplemental Material. Ongoing Clinical Trials Evaluating the Efficacy of GAE in Knee Osteoarthritis

	Registration number	Title	Population	Conditions	Interventions	Characteristics	Dates	Location	Status
1	NCT04472091	Arterial Embolization for Osteoarthritis and Knee Pain	Enrollment: 30 Age: 40 Years and older (Adult, Older Adult)	•Knee Osteoarthritis •Knee Pain Chronic •Knee Pain Swelling •Knee Arthritis •Arterial Occlusion	•Device: Genicular artery embolization (GAE)	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Number of patients with treatment related adverse events •Mean change from baseline pain scores •Mean change from baseline knee function scores	Study Start: Nov 1, 2020 Study Completion: Nov 1, 2024	•Stanford University, Palo Alto, California, USA	Not yet recruiting
2	NCT04456569	Geniculate Artery Embolization for Osteoarthritis	Enrollment: 20 Age: 40 Years to 70 Years (Adult, Older Adult)	•Osteo Arthritis Knee •Arthritis •Osteoarthritis •Osteoarthritis, Knee	•Device: Geniculate Artery Embolization	Study Design: •Allocation: Randomized •Intervention Model: Parallel Assignment •Masking: Single (Outcomes Assessor) •Primary Purpose: Treatment Outcome Measures: •Safety as Assessed by Grade 3-4 Adverse events •Change in Knee Injury and Osteoarthritis Outcome Score (KOOS) •Change in Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) •IL-6 Concentration •Prostaglandin E2 Concentration •Matrix Metalloprotinase 1 Concentration •Vascular Endothelial Growth Factor Concentration •C-Reactive Protein Concentration •Erythrocyte Sedimentation Rate	Study Start: Aug 1, 2020 Study Completion: Feb 2022	•University of Minnesota, Minneapolis, Minnesota, USA	Not yet recruiting
3	NCT04379700	Geniculate Artery Embolization for Treatment of Osteoarthritis	Enrollment: 23 Age: 30 Years to 75 Years (Adult, Older Adult)	•Osteoarthritis	•Procedure: Transcatheter arterial embolization •Device: Embozene™ Microspheres	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Change in Knee pain using Western Ontario and McMaster Universities Osteoarthritis index (WOMAC) score •Change in Knee pain using the Visual Analogue Scale (VAS) scale •Change in Knee Stiffness •Change in Physical Functioning •Change in Quality of life	Study Start: Jan 2021 Study Completion: Jun 2023	Not mentioned	Not yet recruiting

	Registration number	Title	Population	Conditions	Interventions	Characteristics	Dates	Location	Status
4	NCT04113681	Pilot Study: Geniculate Artery Embolization in Knee Osteoarthrosis.	Enrollment: 40 Age: 18 Years and older (Adult, Older Adult)	•Osteoarthritis, Knee	•Procedure: Geniculate Artery Embolization	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Pain control VAS •Function •Radiological examinations	Study Start: Feb 14, 2019 Study Completion: Dec 31, 2021	•CIUSSS de l'Est-de-l'Île-de-Montréal, Installation Hopital Maisonneuve-Rosemont, Montréal, Quebec, Canada	Recruiting
5	NCT03884049	Neovascularization Embolisation for Knee Osteoarthritis.(NEO)	Enrollment: 58 Age: 18 Years and older (Adult, Older Adult)	•Osteoarthritis, Knee	•Procedure: Embolization •Procedure: Sham embolization	Study Design: •Allocation: Randomized •Intervention Model: Parallel Assignment •Masking: Double (Participant, Investigator) •Primary Purpose: Treatment Outcome Measures: •KOOS pain sub score •Total KOOS •painDETECT questionnaire •ICOAP questionnaire •Pain Score 0-10 Numerical Rating Scale (NRS) •EQ-5D-5L questionnaire •Pressure pain threshold testing •Knee MRI	Study Start: Jun 5, 2019 Study Completion: Nov 1, 2021	•ErasmusMC, Rotterdam,	Recruiting
6	NCT03835988	Geniculate Artery Embolization for Knee Osteoarthritis	Enrollment: 20 Age: 21 Years and older (Adult, Older Adult)	•Osteo Arthritis Knee	•Device: Geniculate Artery Embolization	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Patient Pain (mm): visual analog scale •Patient function (arbitrary units) •Medication use (arbitrary units) •Physical function (arbitrary units) •Synovitis (arbitrary units)	Study Start: Mar 1, 2019 Study Completion: Dec31, 2021	•South Healthy Campus, Calgary, Alberta, Canada	Active, not recruiting
7	NCT03491397	Genicular Artery Embolization for the Treatment of Knee Osteoarthritis	Enrollment: 40 Age: 40 Years to 79 Years (Adult, Older Adult)	•Knee Osteoarthritis •Degenerative Joint Disease of Knee	•Device: Embozene MicroSpheres	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Safety evaluated by cumulative rates of Adverse Events •Change in Western Ontarioand McMaster Universities Osteoarthritis Index (WOMAC) as a measure of efficacy •Changes in Visual Analog Scan (VAS) as a measure of efficacy	Study Start: Feb 10, 2019 Study Completion: Jun 1, 2021	•Siddharth Padia, MD, Santa Monica, California, USA	Active, not recruiting

	Registration number	Title	Population	Conditions	Interventions	Characteristics	Dates	Location	Status
8	NCT03460665	Symptomatic Efficacy of Microparticle Arterial Embolization in Knee Osteoarthritis Resistant to Medical Treatment	Enrollment: 90 Age: 40 Years to 80 Years (Adult, Older Adult)	•Knee Osteoarthritis	•Procedure: Arteriography	Study Design: •Allocation: Randomized •Intervention Model: Parallel Assignment •Masking: Double (Participant, Outcomes Assessor) •Primary Purpose: Treatment Outcome Measures: •VAS : Change of pain of the Knee injury •Knee injury and Osteoarthritis Outcome Score (KOOS) •EQ-5D questionnaire •Knee MRI •Knee X-Ray	Study Start: Oct 4, 2018 Study Completion: Dec 1, 2022	•BREUIL, Nice, France	Recruiting
9	NCT03362957	Geniculate Artery Embolization for Knee Pain Secondary to Osteoarthritis (OA)	Enrollment: 21 Age: 40 Years and older (Adult, Older Adult)	•Osteo Arthritis Knee	•Device: Geniculate Artery Embolization •Diagnostic Test: Sham Procedure	Study Design: •Allocation: Randomized •Intervention Model: Crossover Assignment •Masking: Double (Participant, Outcomes Assessor) •Primary Purpose: Treatment Outcome Measures: •Patient function (arbitrary units) •Patient pain (mm) •Reduction in Medication (arbitrary units)	Study Start: May 25, 2018 Study Completion: Nov 2020	•UNC Hospitals, Chapel Hill, North Carolina, USA •Vascular Institute of Virginia, Woodbridge, Virginia, USA	Active, not recruiting
10	NCT02850068	Geniculate Artery Embolization for the Treatment of Knee Pain	Enrollment: 20 Age: 40 Years and older (Adult, Older Adult)	•Osteoarthritis Of Knee	•Device: Geniculate Artery Embolization	Study Design: •Allocation: N/A •Intervention Model: Single Group Assignment •Masking: None (Open Label) •Primary Purpose: Treatment Outcome Measures: •Patient Function (Units on a Scale) •Patient Pain (mm) •Reduction in Medication (Percentage of Participants With a Reduction in Medication Therapy at Month 6)	Study Start: Jan 25, 2017 Study Completion: Aug 3, 2018	•UNC Hospitals, Chapel Hill, North Carolina, USA •Vascular Institute of Virginia, Woodbridge, Virginia, USA	Completed
11	ISRCTN18266598	Geniculate artEry embolisatioN in patiEnts with oSteoarthrItiS of the knee (GENESIS)	Enrollment: 100 Age: 45 Years and older (Adult, Older Adult)	•Osteoarthritis of the knee	•Device: Geniculate Artery Embolization	Study Design: Non-randomised; Both; Design type: Treatment, Imaging, Other, Validation of investigation /therapeutic procedures Outcome Measures: 1. Pain caused by knee osteoarthritis assessed using a visual analogue scale (VAS) at baseline and 6 weeks, 3 months, 1 year and 2 years after the procedure 2. Patients evaluation of their knee symptoms assessed using the Knee injury and Osteoarthritis Outcome Score (KOOS) at baseline and 6 weeks, 3 months, 1 year and 2 years after the procedure	Study Start: Nov 22, 2017 Study Completion: Dec, 2022	•Royal Berkshire Hospital, University of Reading, UK	Active, not recruiting
12	ACTRN12616001184460	Evaluating transcatheter arterial embolisation for improvement of pain in OA of the knee – a randomised controlled trial	Enrollment: 58 Age: 18 Years and older (Adult, Older Adult)	•Knee osteoarthritis	•Device: Geniculate Artery Embolization	Study Design: Randomised controlled trial Outcome Measures: 1. Change in knee pain as assessed by the Knee injury and Osteoarthritis Outcome Score (KOOS) 2. Change in self-reported physical function as measured by the KOOS Function Daily Living scale and KOOS Function, Sports and Recreational Activities scale	Study Start: August, 2017 Study Completion: Aug 2021	•Barwon Health - Geelong Hospital campus,Victoria, Australia	Recruiting